In this program, we build a Vehicle Rental Management System using Java OOP concepts.

First, we create an abstract class named Vehicle. It contains a protected vehicleId field, a constructor, and an abstract calculateRentalCost() method. It also has a concrete printRentalDetails() method to display the vehicle ID, rental duration, and total rental cost.

The Car and Bike classes extend Vehicle and use their daily rates to calculate the rental cost. Jodi rental days 0 ba negative hoy, tahole cost 0 return kore. The Bike class also gives a 10% discount if the rental duration is more than 5 days.

The LuxuryCar class extends Car and reuses the dailyRate field from Car. It also has an extra insuranceFee, which is added to the rental cost. Invalid rental duration hole etao 0 return kore.

Each subclass overrides the calculateRentalCost() method according to its own rules. This demonstrates method overriding.

In the Main class, we create a Vehicle[] array containing one Car, one Bike, and one LuxuryCar object. Then, we use a loop to calculate the rental cost and print the details of each vehicle. Java automatically calls the correct overridden method for each object. This demonstrates runtime polymorphism.

We also include an invalid rental duration to check whether the program correctly returns 0 for an invalid number of rental days.